HOCHSCHULE
HAMM-LIPPSTADT

**Stefan Henkler** E-Mail: stefan.henkler@hshl.de

**Faezeh Pasandideh** E-Mail: faezeh.pasandideh@hshl.de

# Boeing 737 MAX-Failed

HOCHSCHULE HAMM-LIPPSTADT

*What happens when a hardware problem gets a software band-aid*

**THE PROBLEM**

**Bigger engines**
wouldn't fit the old frame.

Boeing moved them
forward & higher.
Nose tilted up — dangerous.

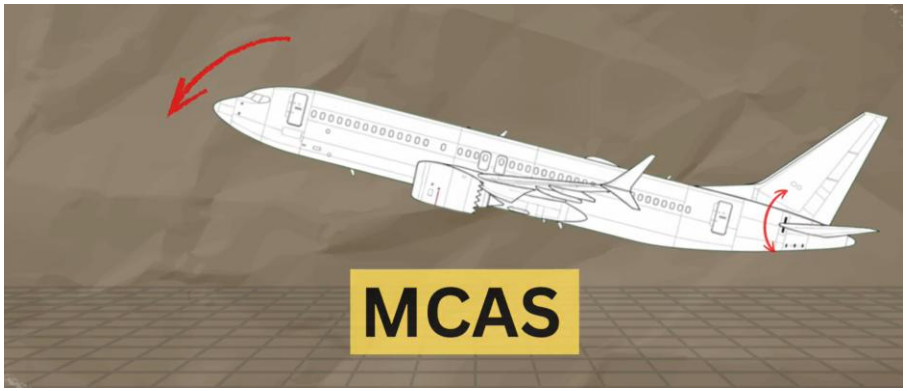
**THE "FIX"**

Boeing wrote software called **MCAS** to push the nose down.

**Pilots never told**
**it existed.**
Used only ONE sensor.

**THE RESULT**

Faulty sensor fired.
MCAS pushed down.
Pilots fought back.
MCAS pushed again.
**No way to stop it.**
**346 people killed in 2 crashes.**






# Development Live Cycle

- International Council on Systems Engineering (INCOSE Style)
- Life cycle processes

HOCHSCHULE HAMM-LIPPSTADT

# Systems Engineering

1. Introduction
2. Definitions
3. Generic Life-Cycle
4. Technical Process
5. Project Process
6. Agreement Process
7. Organizational Process
8. Tailoring Process

- In this lecture
  - Systems Engineering based on INCOSE Systems Engineering Handbook
    - INCOSE Systems Engineering Handbook v. 4.0, October 2015, INCOSE-TP-2003-002-03.2.2
    - https://www.sebokwiki.org/wiki/INCOSE_Systems_Engineering_Handbook
  - Why?
    - Learn standardizes systems engineering practice
      - Handbook is consistent with ISO/IEC 15288:2008 – Systems and software engineering – System life cycle processes
- INCOSE:
  - International Council on Systems Engineering
  - champions the art, science, discipline, and practice of systems engineering
    - evolving Systems Engineering Body of Knowledge
      - E.g. provide a description of key process activities performed by systems engineers
    - Systems Engineering Professional (SEP) Certification program

INCOSE The International Council on Systems Engineering (INCOSE) is a not-for-profit membership organization founded to develop and disseminate the interdisciplinary principles and practices that enable the realization of successful systems.

| | |
|---|---|
| 1990 | National Council on Systems Engineering (NCOSE) founded, USA |
| | The goal: Development of Systems Engineerings in the United States |
| 1995 | Extension of the goals to international realm |
| | Re-naming in International Council on Systems Engineering (INCOSE) |
| 1997 | Gesellschaft für Systems Engineering (GfSE) the german branch is founded |
| 1998 | Systems Modeling Language (SysML) |

Today INCOSE has over 9000 members.

INCOSE goals:

- Definition
- Collaboration in Education
- Promotion
- Application

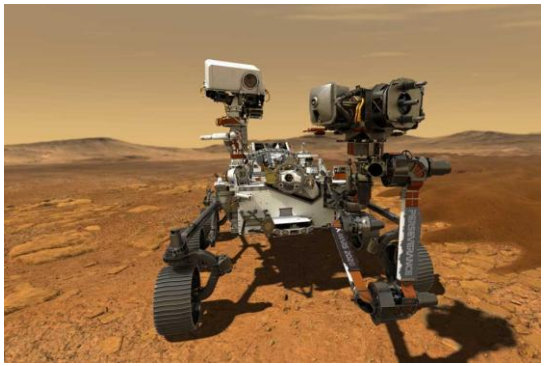
Of Systems Engineering

**INCOSE-Themen:**

- Leadership in SE
- Research
- Education
- Certification

# 7 INCOSE: examples

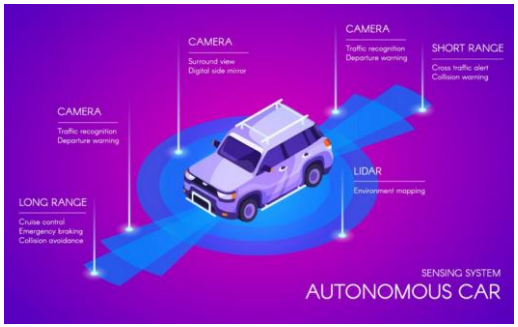
- NASA uses INCOSE methods to design Mars rovers.

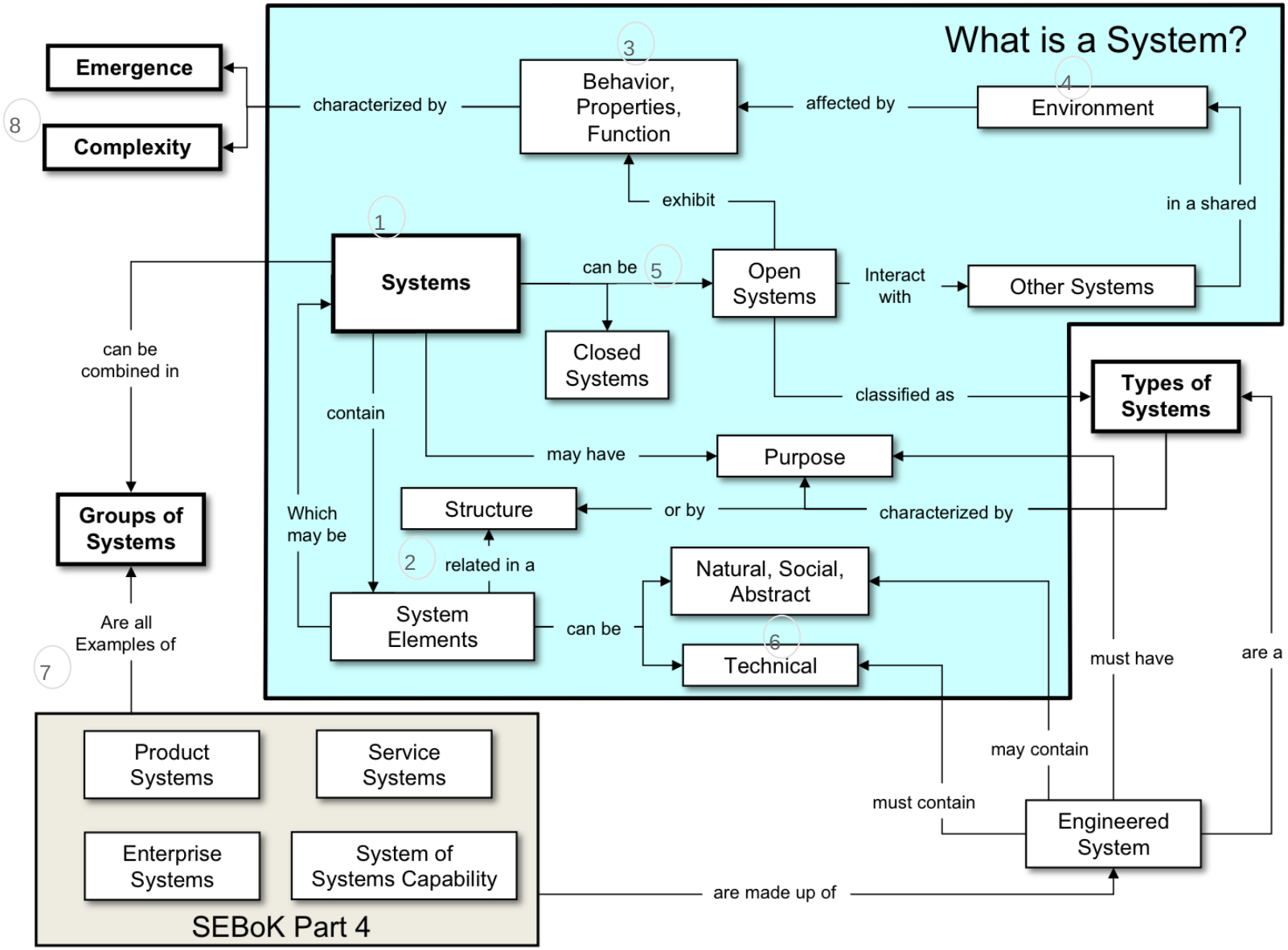


- Airbus uses them to design the A380.

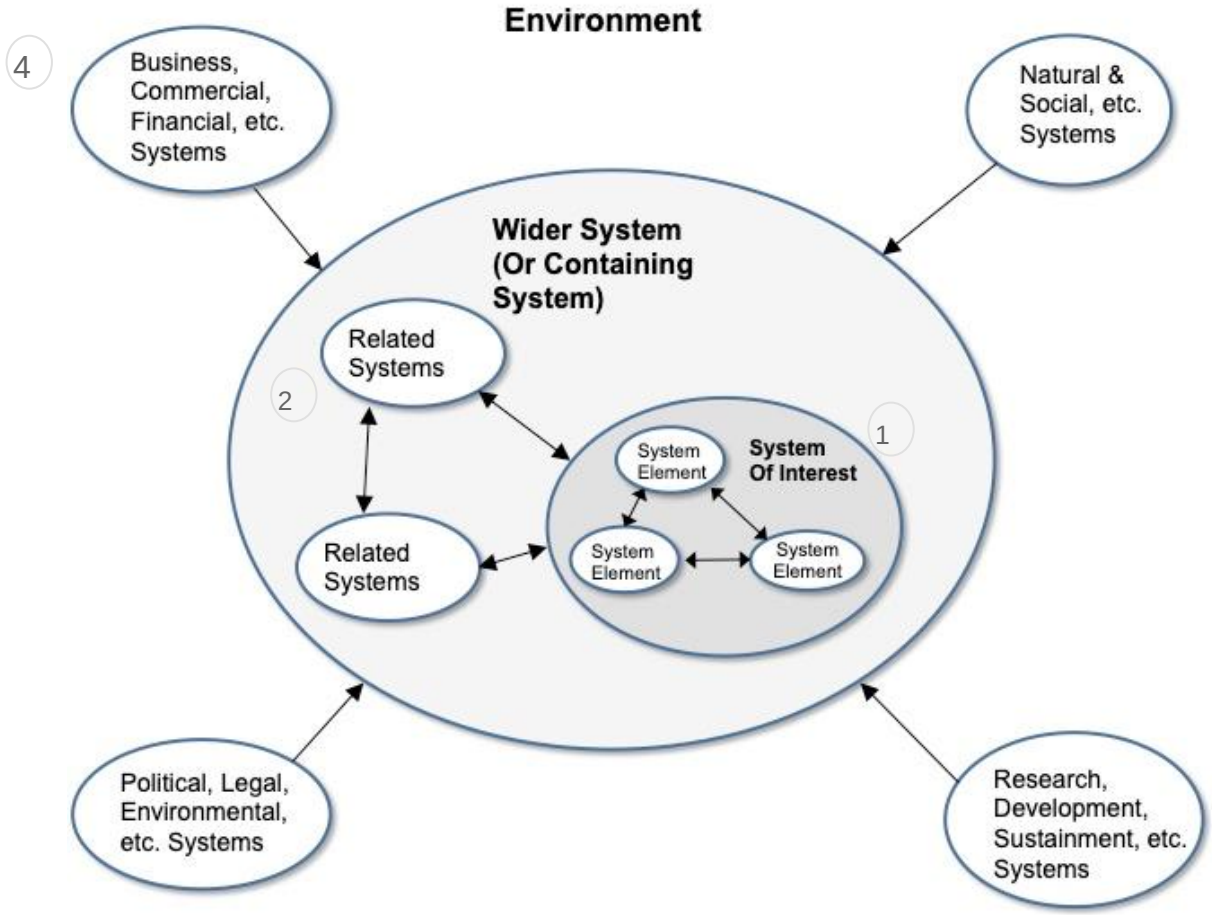



- The German automotive industry uses them for developing autonomous vehicles.




What is a System?
Emergence
Complexity
8
characterized by
3
Behavior, Properties, Function
affected by
4
Environment
exhibit
in a shared
1
Systems
can be
5
Open Systems
Interact with
Other Systems
Closed Systems
can be combined in
contain
classified as
Types of Systems
may have
Purpose
Groups of Systems
Which may be
Structure
or by
characterized by
2
related in a
Natural, Social, Abstract
System Elements
can be
6
Technical
Are all Examples of
7
must have
are a
may contain
must contain
Engineered System
Product Systems
Service Systems
Enterprise Systems
System of Systems Capability
SEBoK Part 4
are made up of

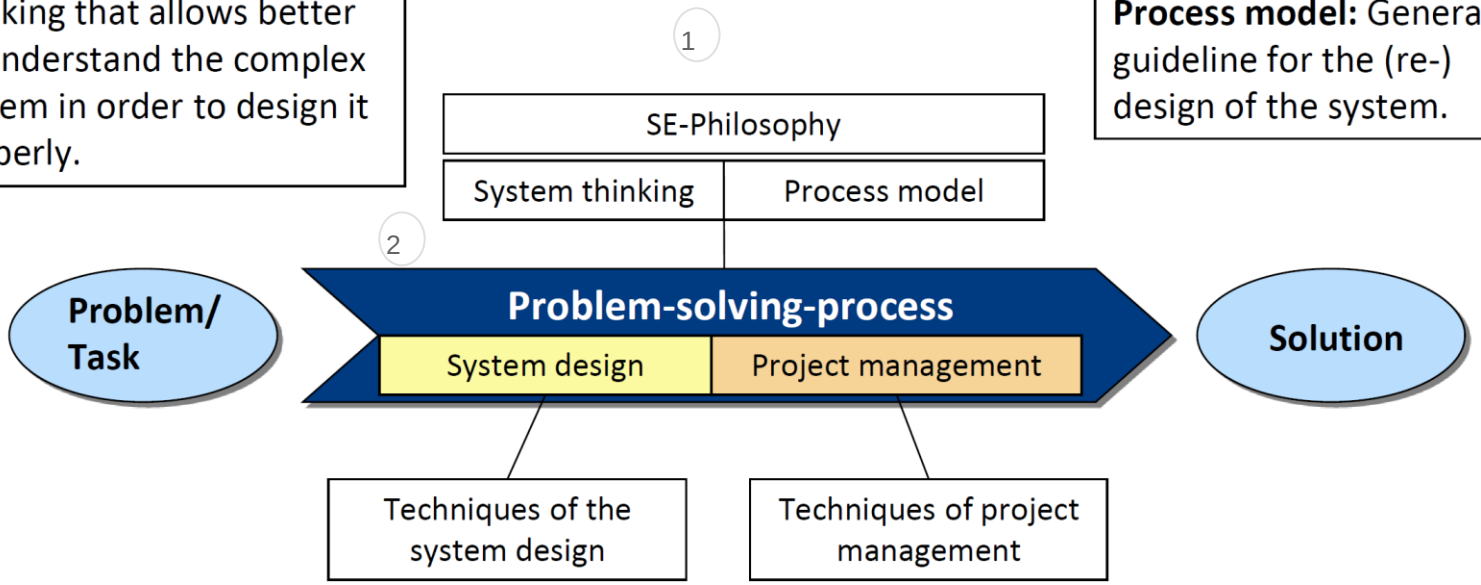
## environment



# 10 Parts of Systems Engineering

**System thinking:** Way of thinking that allows better to understand the complex system in order to design it properly.

**Process model:** General guideline for the (re-) design of the system.



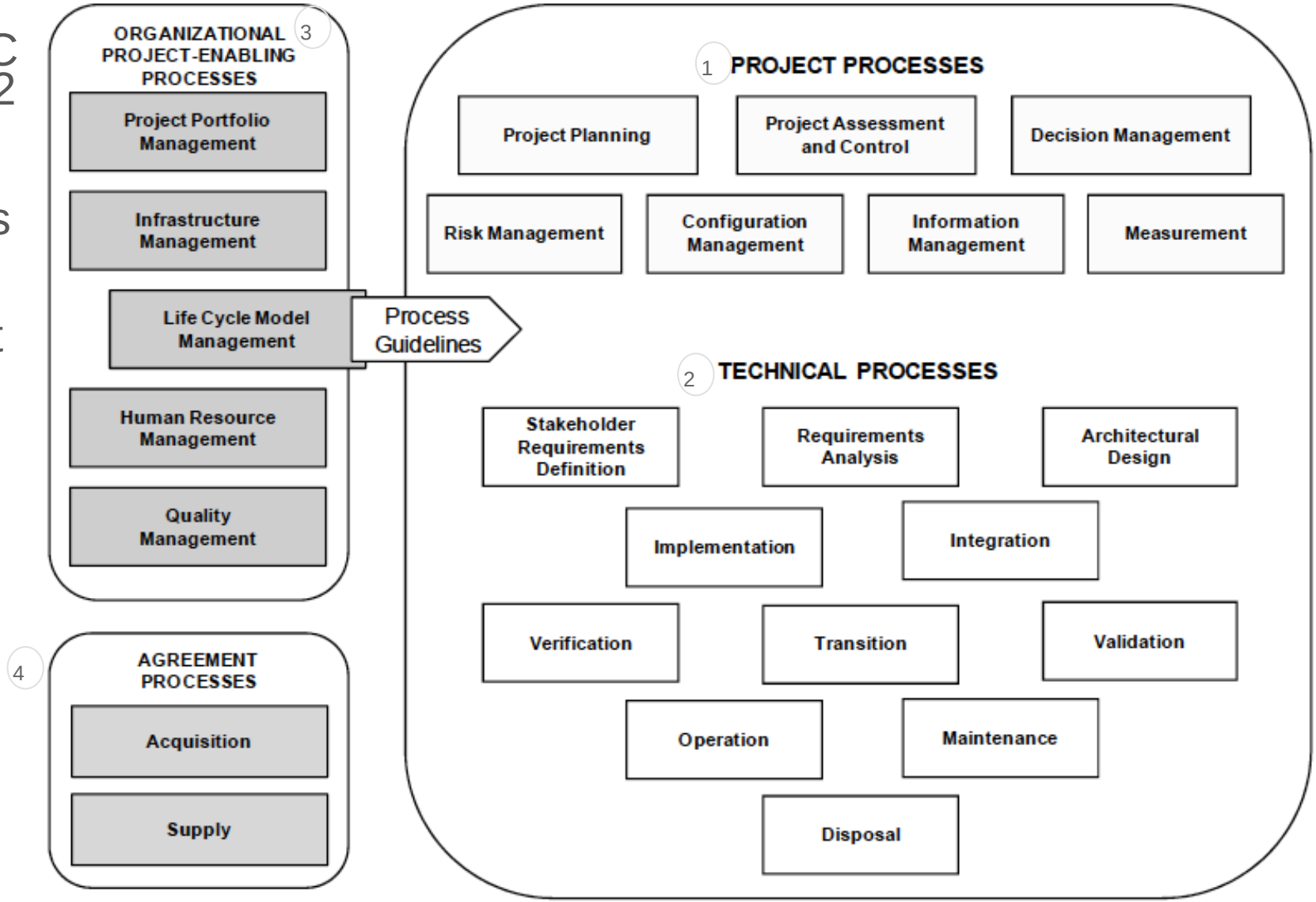

- The **System design** constructive work to find the solution.
- The **Project management**, organization and the coordination of the problem solving process.

**System Design**
- Defines what to build
- Technical architecture
- System integration
- Engineering creativity

**Project Management**
- Defines how to build it
- Time, cost, resources
- Risk management
- Team coordination

# Overview – Systems Life-Cycle Process

- ISO/IEC 15288:2008
- Four process groups to support SE



## Systems Engineering according to INCOSE

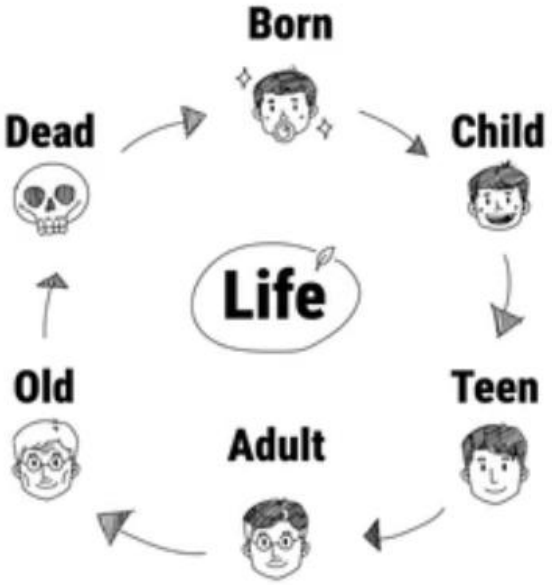
- Systems engineering is an **interdisciplinary** approach and means to **enable** the **realization** of **successful systems**. It focuses **on defining customer needs** and **required functionality early in the development cycle**, documenting requirements, and then proceeding with **design synthesis and system validation** while considering the complete problem: operations, cost and schedule, performance, training and support, test, manufacturing, and disposal. SE considers both the **business** and the **technical needs** of all customers with the goal of providing a quality product that meets the user needs.

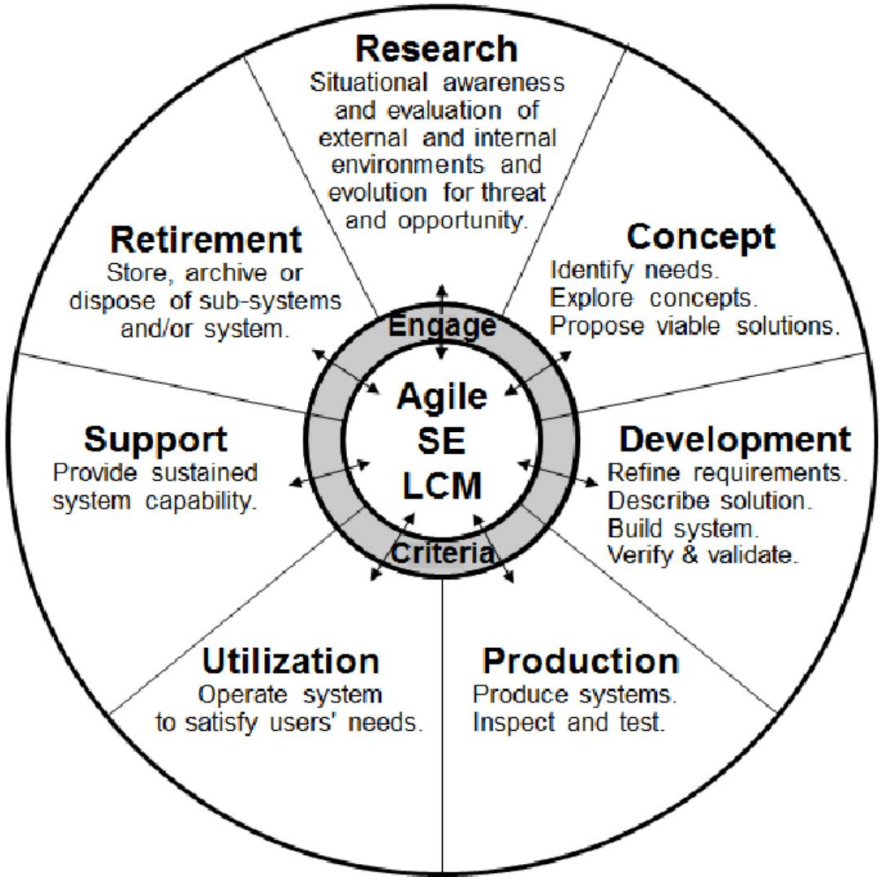
## Life Cycle

- Every **system** has a **life cycle**. … A system progresses through its life cycle as the result of **actions**, **performed** and **managed by people** in organizations, **using processes** for execution of these actions
  - The life cycle model comprises one or more **stage** models, as needed. It is assembled as a sequence of stages that may overlap and/or iterate, as appropriate for the system-of-interest's scope, magnitude, complexity, changing needs and opportunities
- **Life cycles** vary according to the nature, purpose, use and prevailing circumstances of the system. Each stage has a distinct purpose and contribution to the whole life cycle and is conserved when planning and executing the system life cycle. … The stages thus provide organizations with a framework within which organization management has high-level visibility and control of project and Technical Processes

# II.2 Definitions

## Life Cycle



# II.3 Generic Life-Cycle

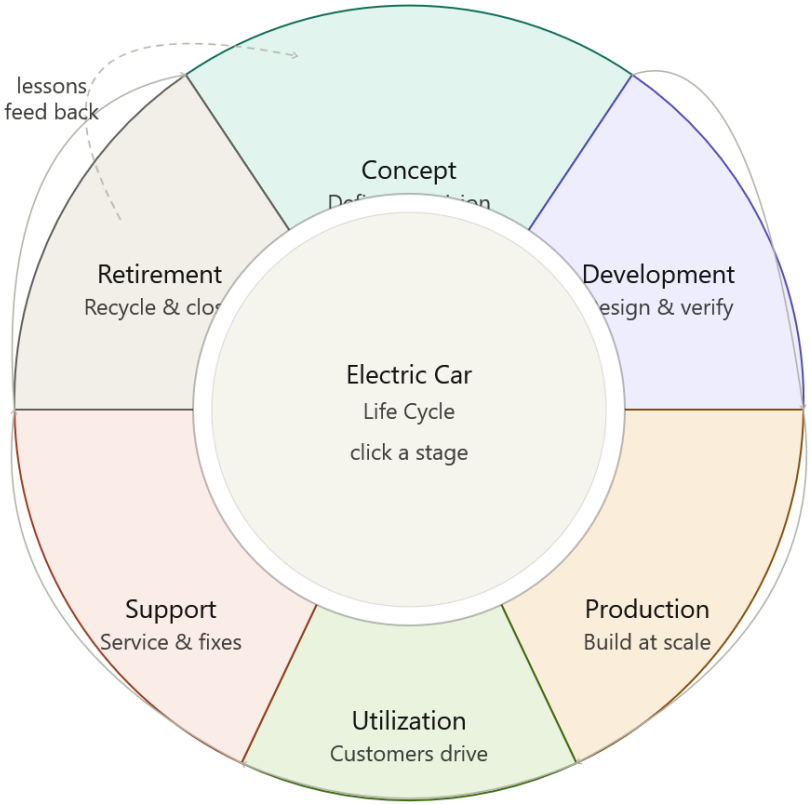
## stages, their purposes, and decision gate options

| Life cycle stages | Purpose | Decision gates |
|---|---|---|
| **Concept** | Define problem space<br>1. Exploratory research<br>2. Concept selection | Decision options<br>• Proceed with next stage<br>• Proceed and respond to action items<br>• Continue this stage<br>• Return to preceding stage<br>• Put a hold on project activity<br>• Terminate project |
| | Characterize solution space | |
| | Identify stakeholders' needs | |
| | Explore ideas and technologies | |
| | Refine stakeholders' needs | |
| | Explore feasible concepts | |
| | Propose viable solutions | |
| **Development** | Define/refine system requirements | |
| | Create solution description—architecture and design | |
| | Implement initial system | |
| | Integrate, verify, and validate system | |
| **Production** | Produce systems | |
| | Inspect and verify | |
| **Utilization** | Operate system to satisfy users' needs | |
| **Support** | Provide sustained system capability | |
| **Retirement** | Store, archive, or dispose of the system | |

[INCOSE2015]

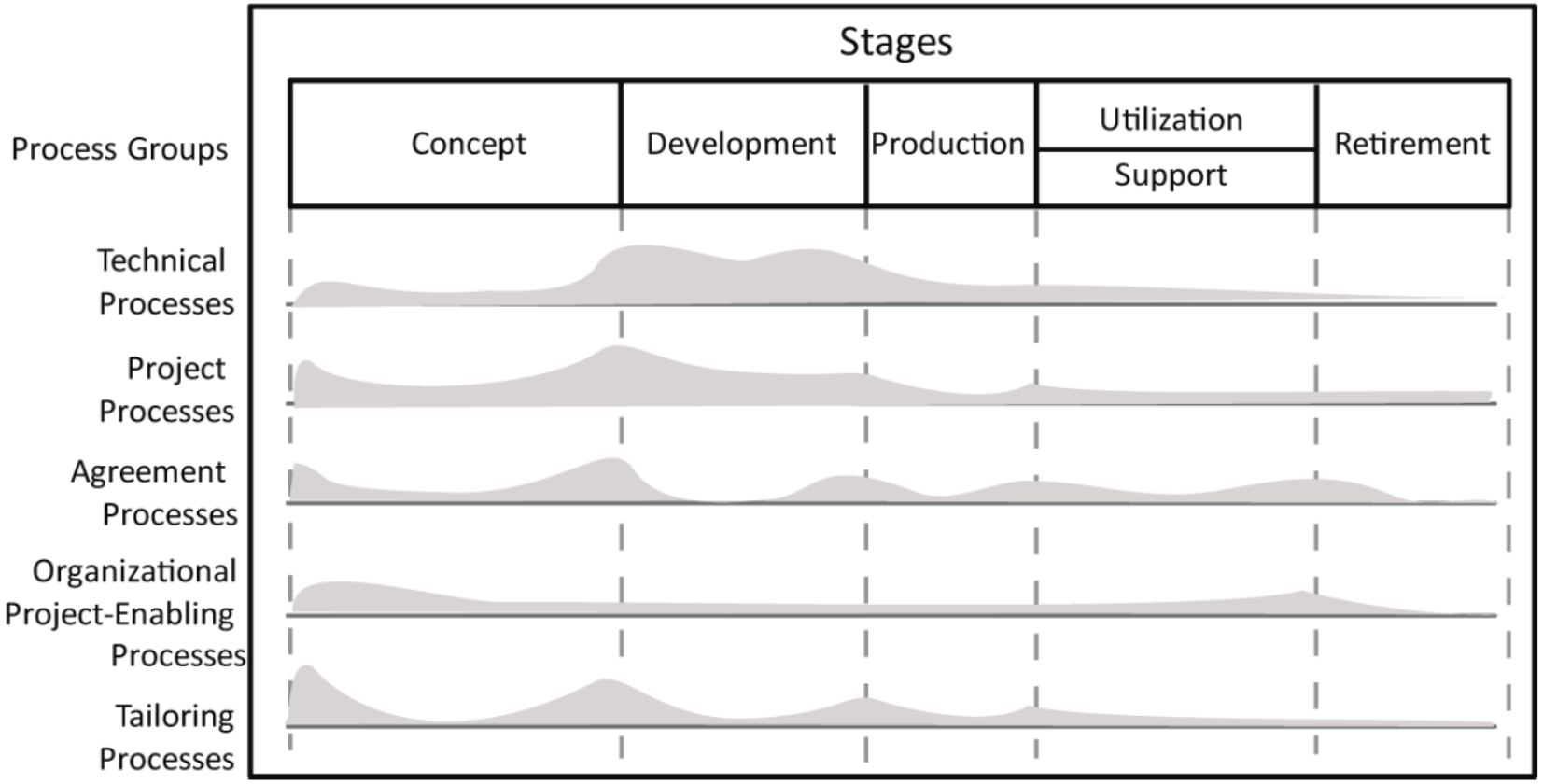
# II.3 Generic Life-Cycle: new electric car

- **Concept**: 'We need a 400km-range EV for under 30,000 euros.'
- **Development**: engineers design battery, motor, software.
- **Production**: factory builds cars.
- **Utilization**: customers drive them.
- **Support**: dealerships service them.
- **Retirement**: recycling programs for batteries..

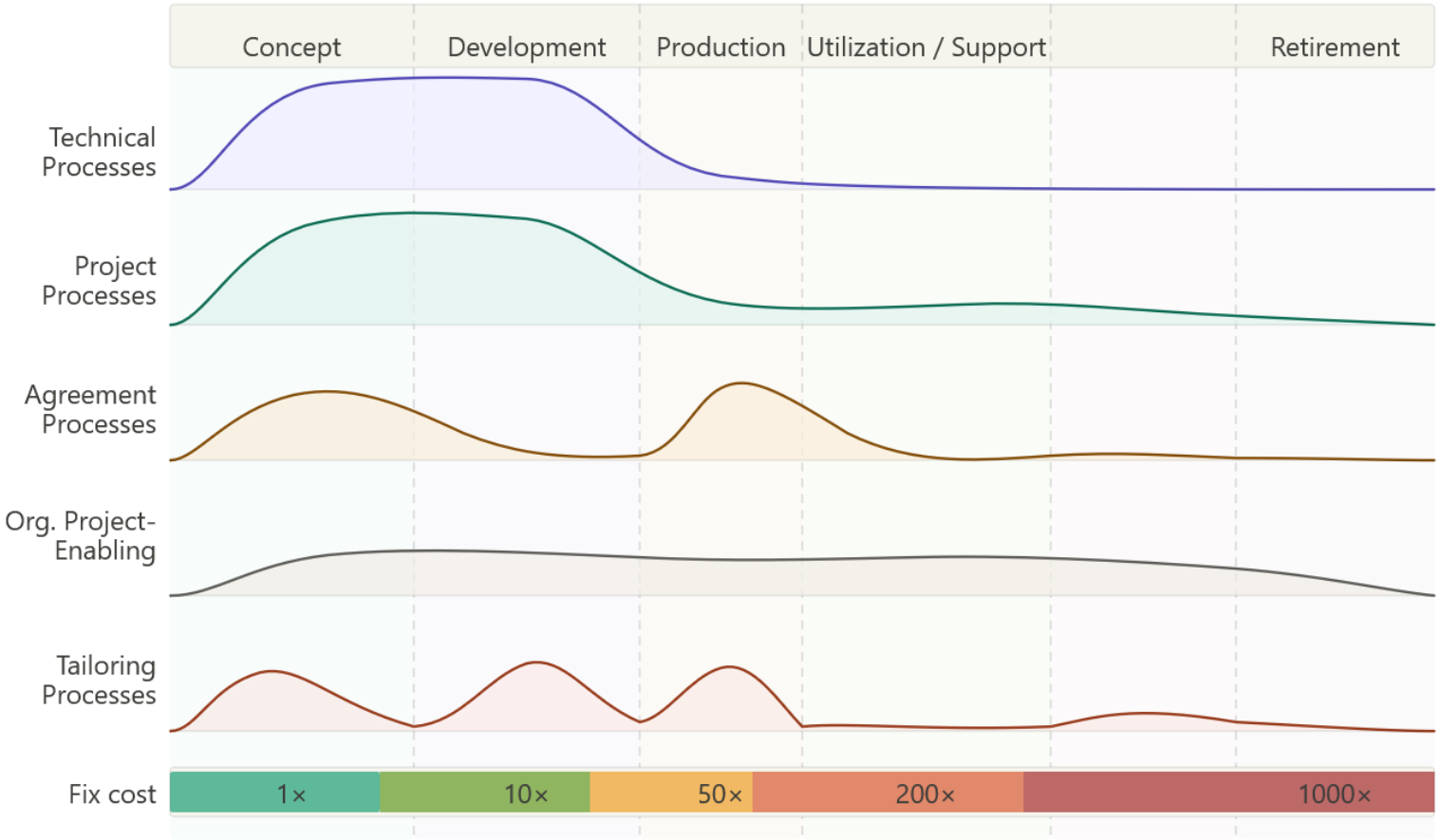


# Generic Live-Cycle

## SE Level of Effort



# Generic Live-Cycle: HSUV



# Generic Life-Cycle

## Comparison of life-cycle models

HOCHSCHULE HAMM-LIPPSTADT

**Generic Life Cycle (ISO 15288:2002)**

| Concept Stage | Development Stage | Production Stage | Utilization Stage / Support Stage | Retirement Stage |
|---|---|---|---|---|

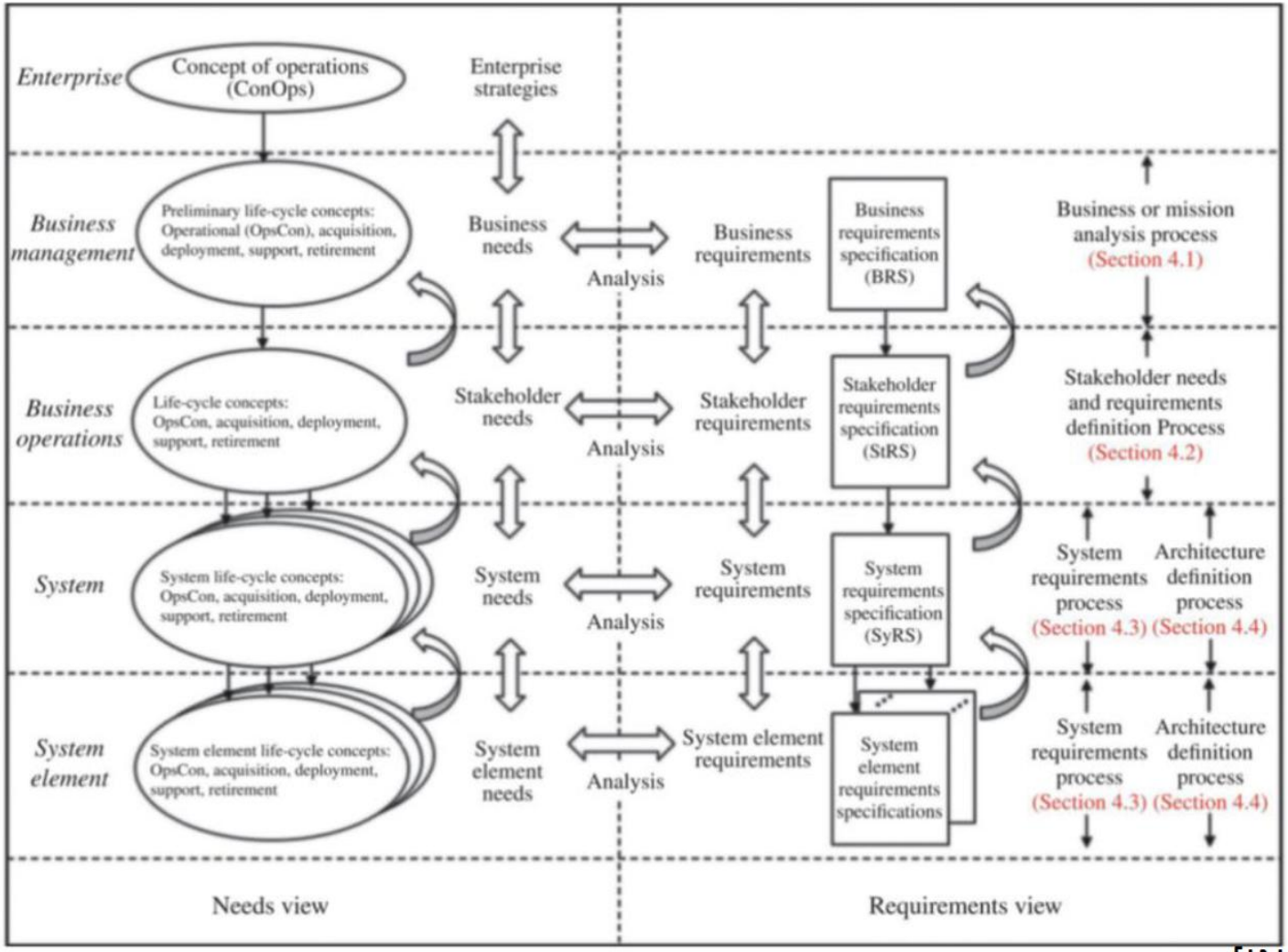
**Typical High-Tech Commercial Systems Integrator**

| Study Period | | | | Implementation Period | | | Operations Period | | |
|---|---|---|---|---|---|---|---|---|---|
| User Requirements Definition Phase | Concept Definition Phase | System Specification Phase | Acq Prep Phase | Source Select. Phase | Development Phase | Verification Phase | Deployment Phase | Operations and Maintenance Phase | Deactivation Phase |

# II.4. Technical Process

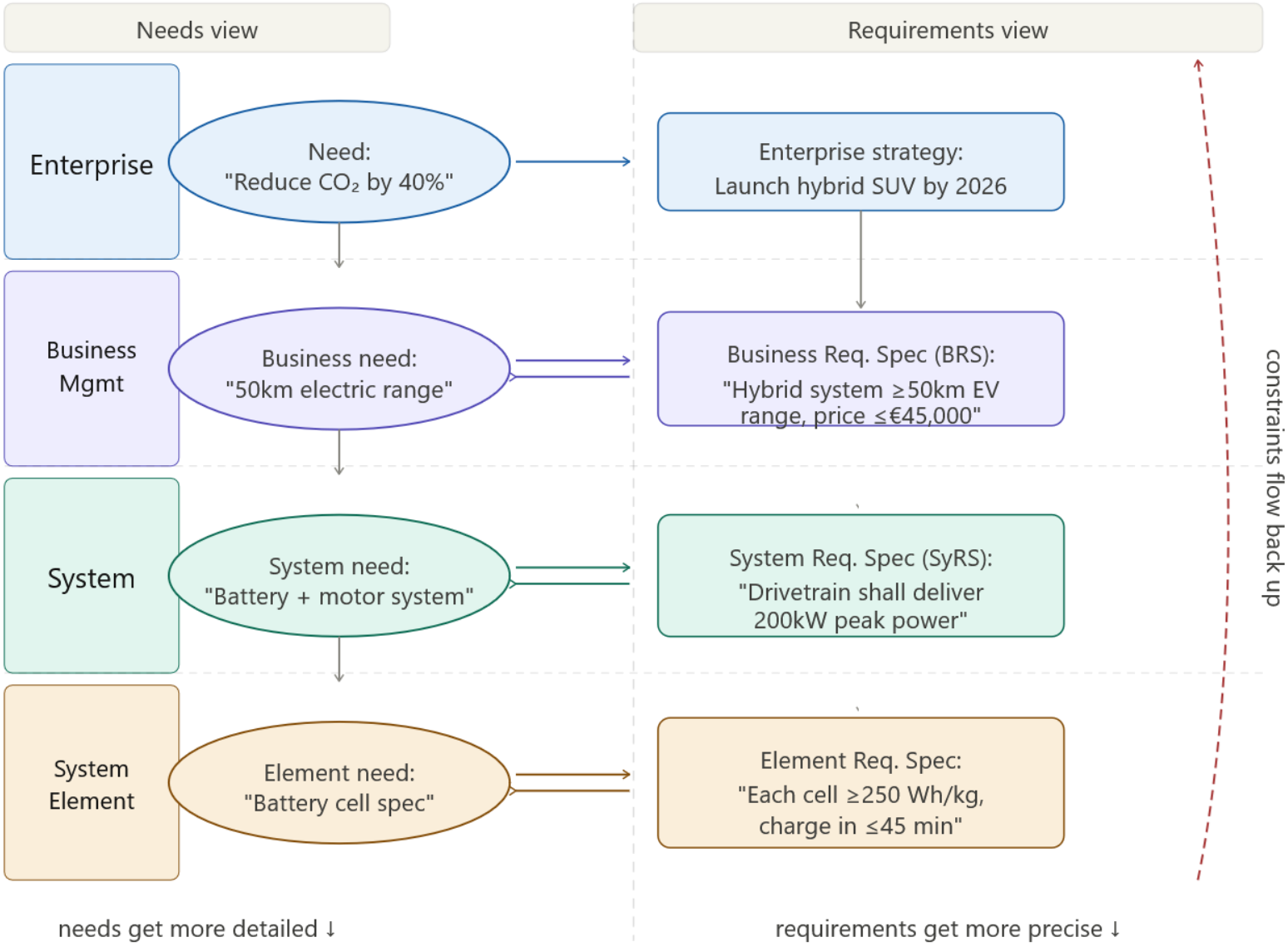
- The **Technical Processes** are used to **define** the **requirements** for a **system**, to **transform** the requirements into an **effective product**, to permit consistent **reproduction** of the product where necessary, to use the product to **provide the required services**, to sustain the provision of those services and to dispose of the product when it is **retired** from service.

# From needs to requirements



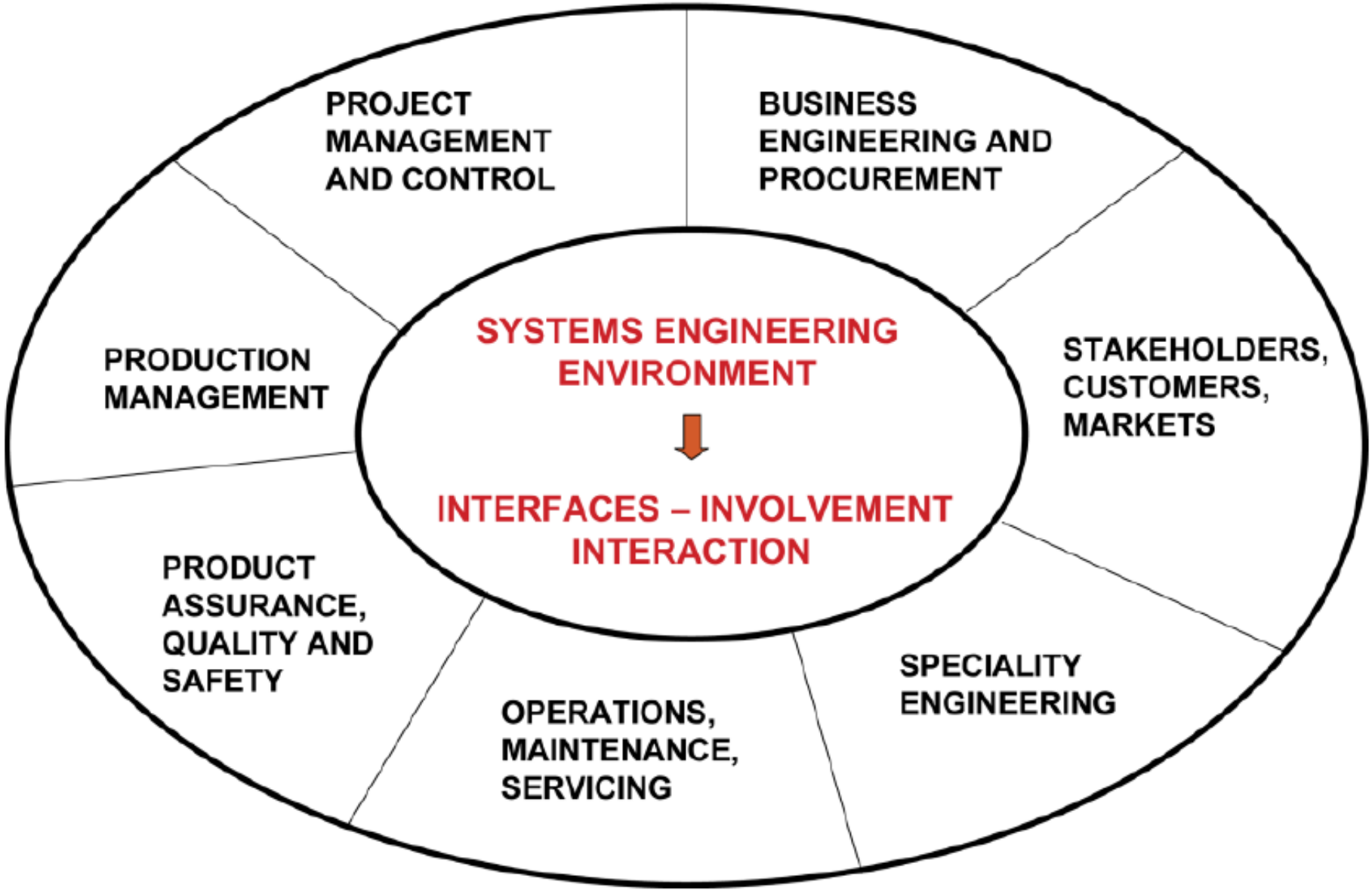

[INCOSE2015]

# From needs to requirements: HSUV



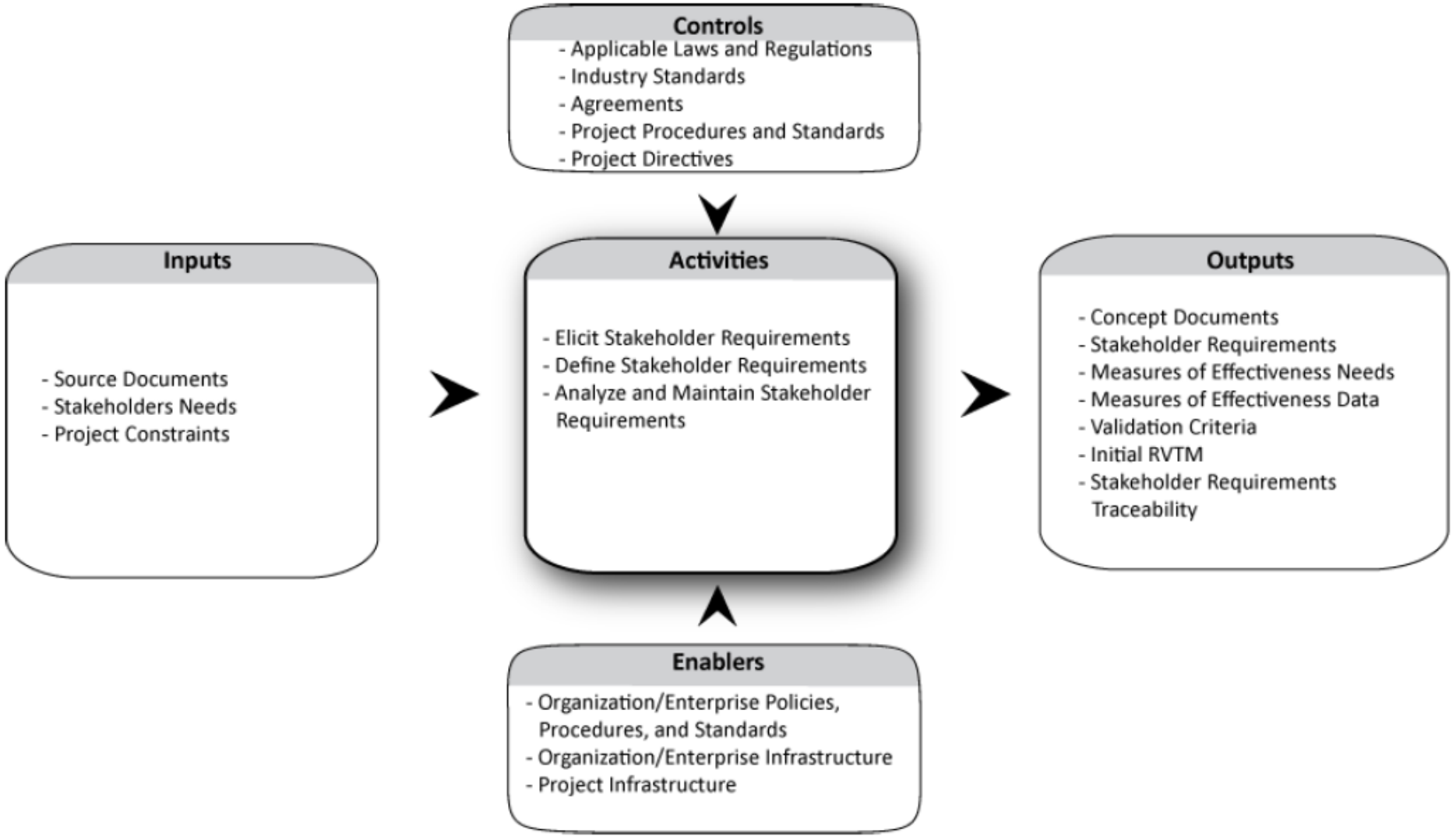
# Technical Process

## Critical Interactions for Systems Engineers



# Technical Process

## Stakeholder Requirements Definition: : WHO needs WHAT, and WHY?

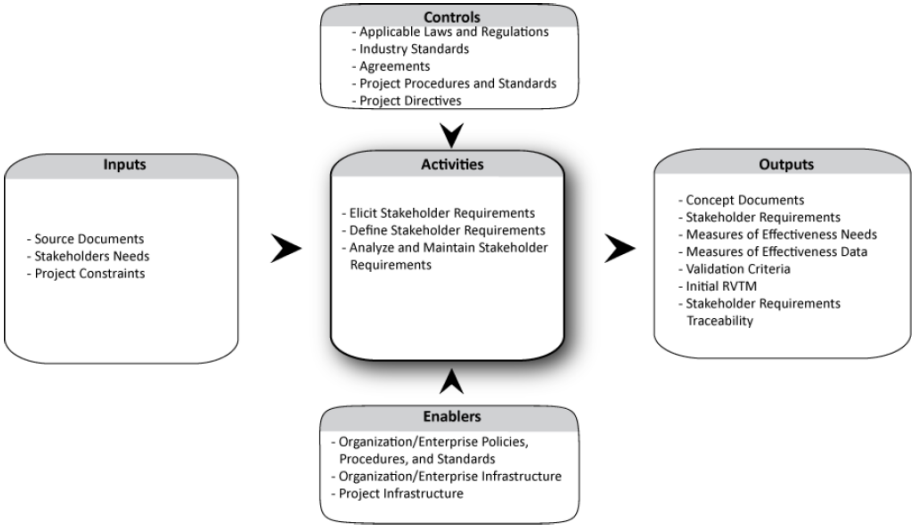



RVTM:Requirements Verification Traceability Matrix

# Technical Process

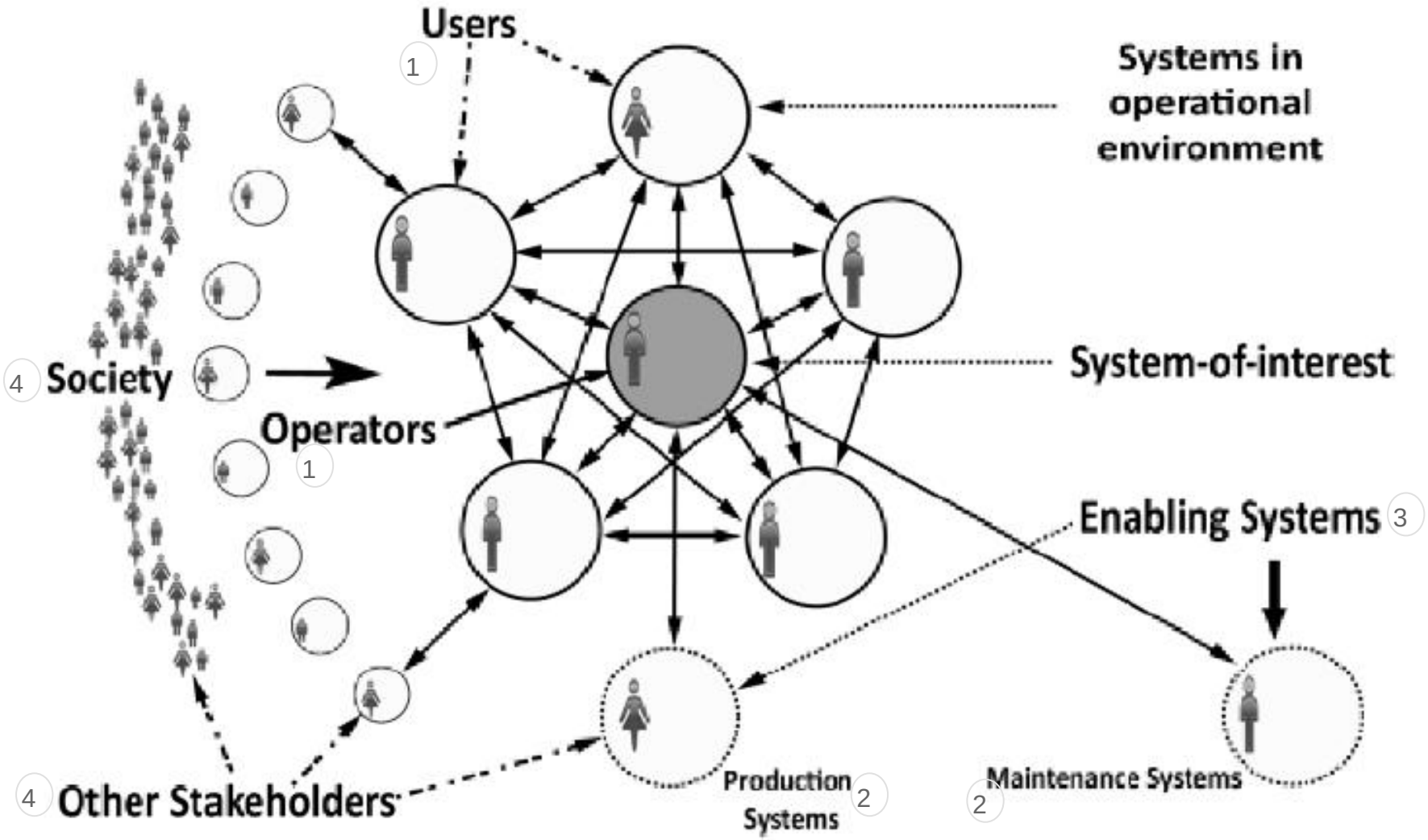
## Stakeholder Requirements Definition: Hsuv

- **Input** — customer wishes, budget, EU laws
- **Controls** — safety standards, regulations, contracts
- **Activities** — elicit, write, update requirements
- **Enablers** — tools, software, trained engineers
- **Output** — formal spec: "≥50km EV range, ≤€45,000"



# Stakeholder Requirements Definition

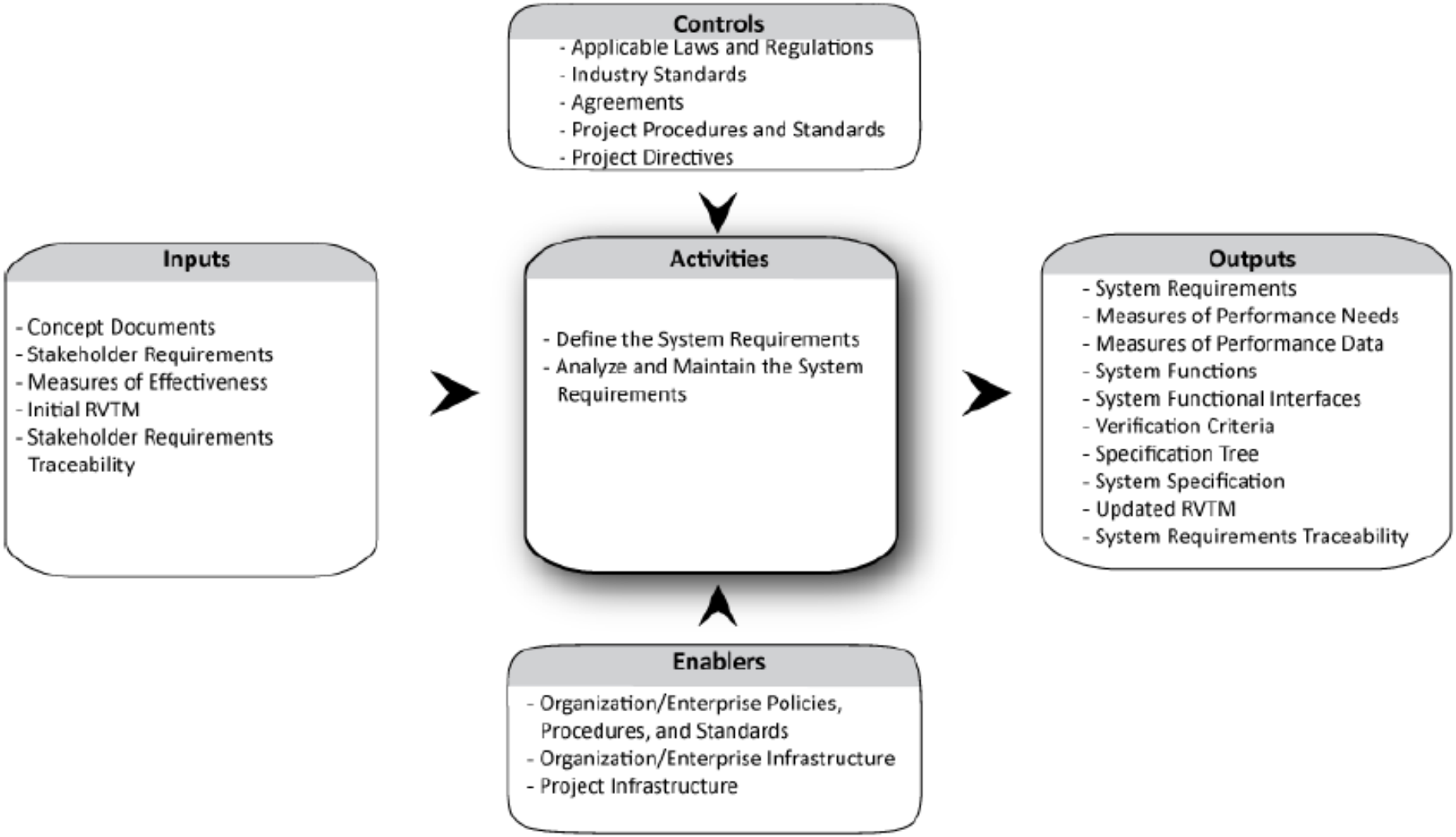
## Identify Users and Stakeholder



## Requirements Analysis Process

- The purpose of the Requirements Analysis Process is to **transform the stakeholder**, requirement-driven view of desired services **into a technical view** of a **required product** that could **deliver** those services.

# Requirements Analysis Process



# Requirements Analysis Process:SUV

- **Input** — customer wishlist + concept sketches from previous step: "50km range, under €45,000“
- **Activity 1:** Define — translate wishes into engineering language: "battery ≥15kWh, motor ≥80kW“
- **Activity 2:** Analyze — check for contradictions, gaps, and untestable requirements
- **Output 1:** System Specification — precise, measurable, buildable requirements in engineering language
- **Output 2:** RVTM — one row per requirement, last column = which test proves it was met

## Characteristics of Good Requirements

- (it must be) necessary
- Implementation independent
- Clear and concise
- Complete
- Consistent
- Achievable
- Traceable
- Verifiable

# Requirements Analysis Process:SUV

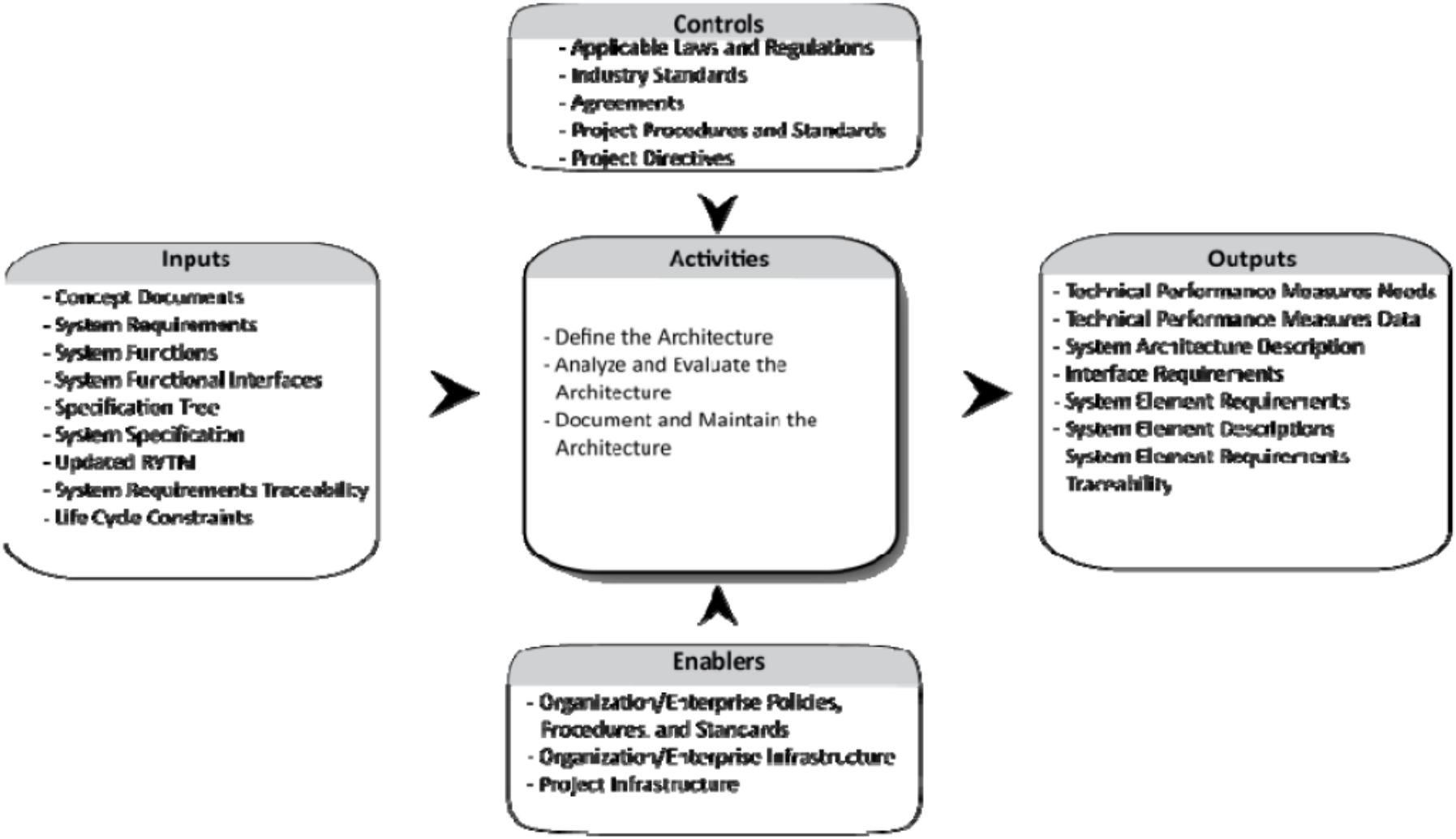
## Characteristics of Good Requirements

HOCHSCHULE HAMM-LIPPSTADT

| Property | Meaning | Bad requirement | Good requirement |
|---|---|---|---|
| **Necessary** | Must exist for a real reason | The SUV shall have a green charging indicator light | The SUV shall notify the driver when battery drops below 20% |
| **Implementation Independent** | Says what, not how | The SUV shall use a 400V lithium-NMC battery pack | The SUV shall store ≥15kWh of usable electrical energy |
| **Clear and Concise** | One meaning, no ambiguity | The SUV shall switch between modes smoothly and quickly | Mode transition from EV to hybrid shall complete within 500ms |
| **Complete** | All information included | The electric motor shall be powerful | The electric motor shall deliver ≥80kW continuous power at 20°C ambient |
| **Consistent** | No contradictions | The SUV shall prioritize EV mode. The SUV shall prioritize fuel economy. | The SUV shall operate in EV mode by default when battery is above 30% |
| **Achievable** | Technically and financially possible | The battery shall last 20 years under daily fast charging | The battery shall retain ≥80% capacity after 1,500 full charge cycles |
| **Traceable** | Linked to a real stakeholder need | (no source given) | REQ-047: derived from Stakeholder Need SN-012 — "customer wants low city fuel cost" |
| **Verifiable** | Can be tested or measured | The SUV shall have good electric range | The SUV shall achieve ≥50km EV range under WLTP Class 3 test conditions |

# Technical Process

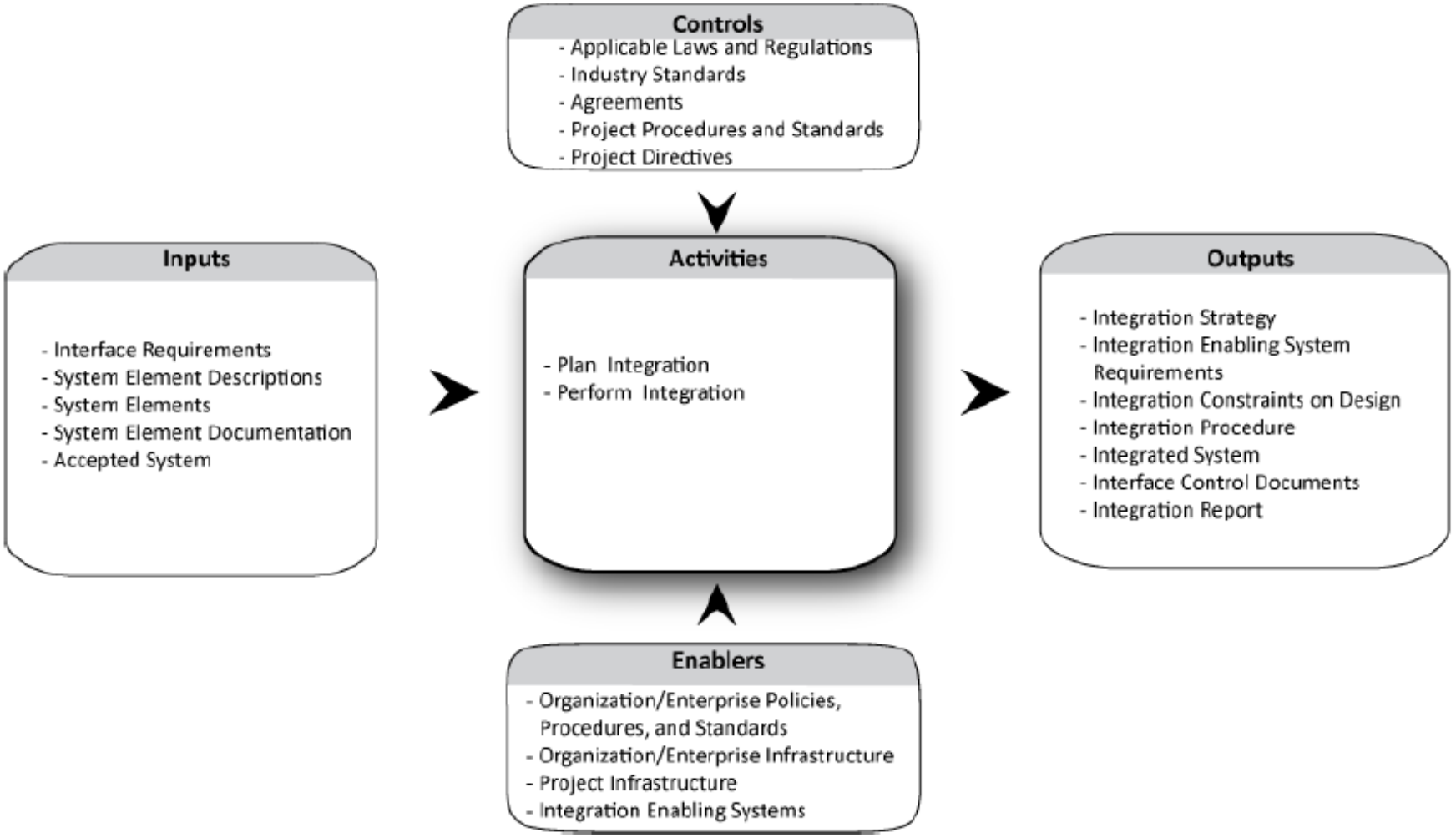
## Architectural Design Process



# Technical Process

## Integration Process

- *The purpose of the Integration Process is to assemble a system that is consistent with the architectural design.*

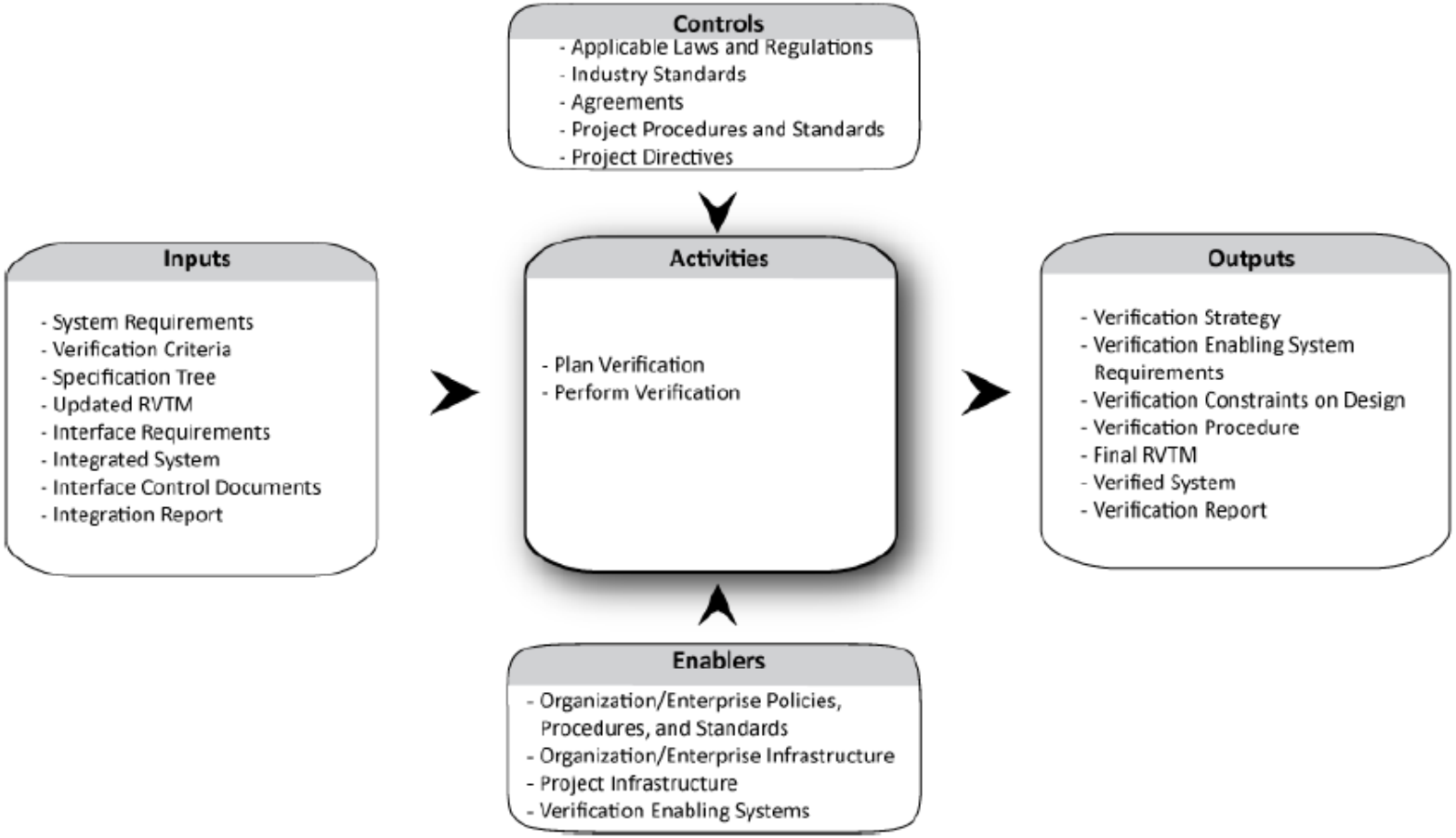


## Integration Process: HSUV

- **Inputs** : approved specs, architecture docs, component requirements: everything the factory needs to start building.
- **Controls**: EU laws, ISO 26262, BMW quality standards: the rulebook nobody on the factory floor can ignore.
- **Activities**:
  - Plan: who builds what, in what order, on which line
  - Perform: robots weld, technicians install the battery pack, software is flashed onto the ECU
- **Enablers:** the Dingolfing factory, battery test rigs, dynamometer, PLM tracking software: everything that must exist before building can start.
- **Outputs**: not just the car, but also:
  - The physical components: battery pack, motor, power electronics
  - As-built documentation for every part
  - Trained dealership technicians ready to service it for 15 years

## Verification Process

- *The purpose of the Verification Process is to confirm that the specified design requirements are fulfilled by the system.*



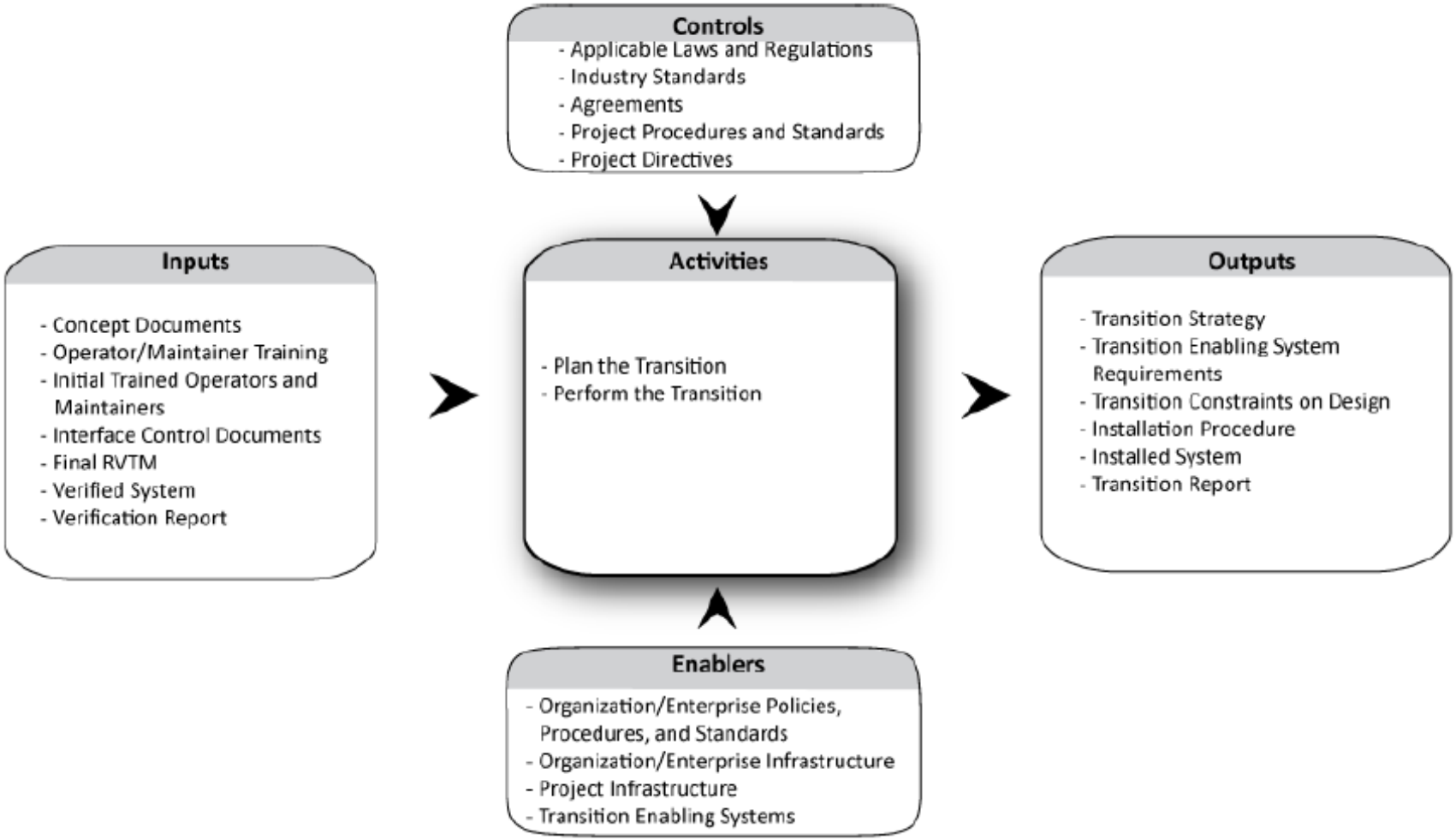
## Verification Process

- **Inputs** — assembled SUV, requirements list, RVTM
- **Controls** — EU laws, ISO 26262, WLTP test protocol
- **Activities**:
  - Plan: match each requirement to a test
  - Perform: dyno, crash, -20°C chamber, brake test
- **Enablers** — Miramas proving ground, crash facility, dynamometer, RVTM software

- **Outputs**:
  - Verified SUV with formal sign-off
  - Final RVTM — every requirement ticked
  - Verification report → submitted for EU type approval
  - Failed tests → flagged back to design team

## Transition Process

- *The purpose of the Transition Process is to establish a capability to provide services specified by stakeholder requirements in the operational environment.*



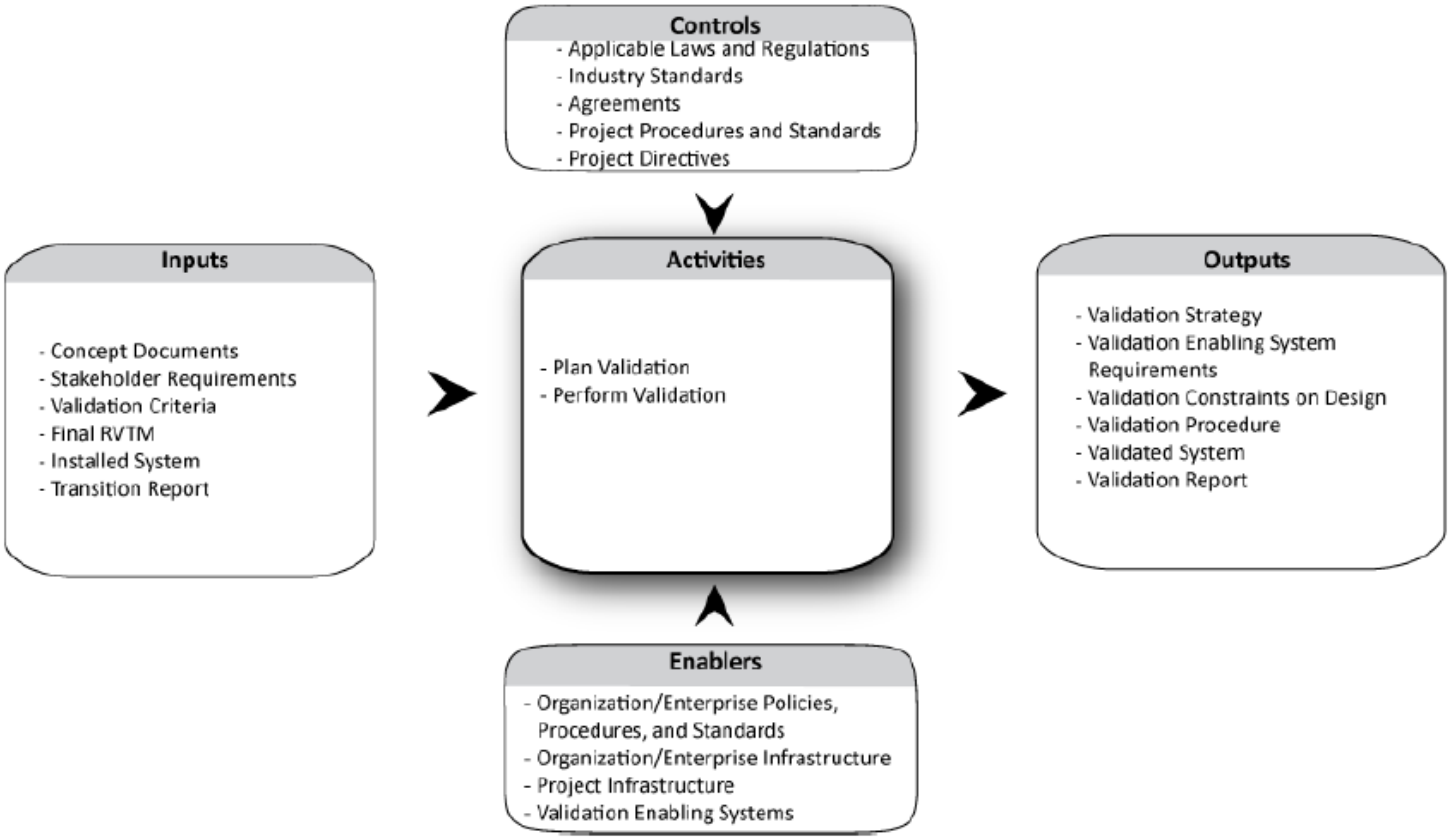
## Transition Process:HSUV

SUV passed all tests — now hand it over to dealerships and customers

- **Inputs** — verified SUV, verification report, final RVTM, trained technicians
- **Controls** — EU regulations, BMW dealer agreements, homologation certificates
- **Activities**:
  - Plan: which dealerships, which models, in what order
  - Perform: ship cars, install charging equipment, brief service teams
- **Enablers** — BMW dealer network, logistics, training centres
- **Outputs**:
  - SUV delivered to dealership, ready for customer
  - Installation procedure for dealership setup
  - Transition report confirming handover complete
  - Issues found → flagged back to factory

## Validation Process

- *The purpose of the Validation Process is to provide objective evidence that the services provided by a system when in use comply with stakeholders' requirements, achieving its intended use in its intended operational environment.*



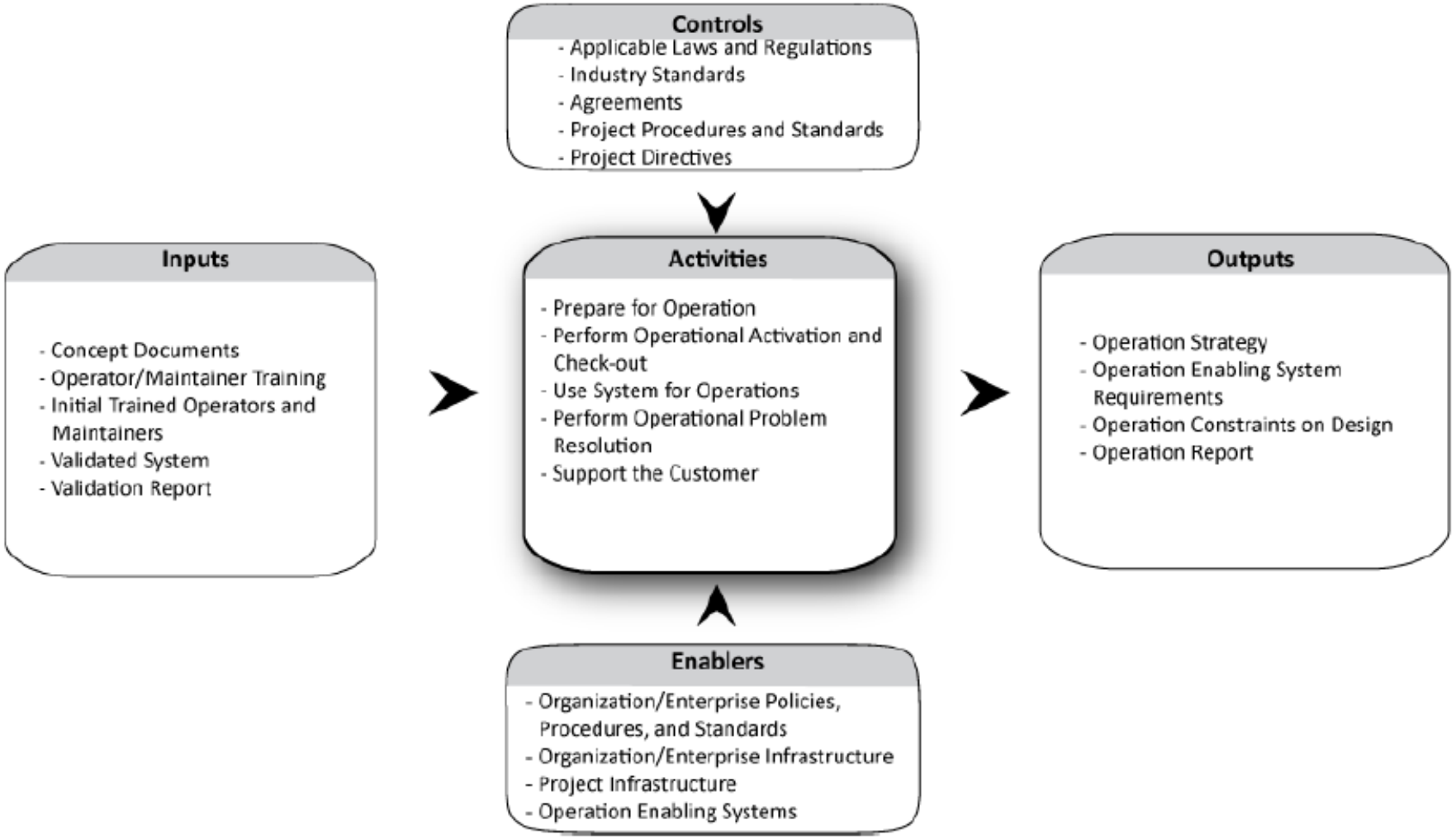
## Validation Process: HSUV

**Did we build the *right* thing? Real customers, real roads — not a lab.**

- **Inputs** — installed SUV, original stakeholder requirements, validation criteria, transition report
- **Controls** — EU consumer laws, BMW acceptance standards, real-world driving protocols
- **Activities**:
  - Plan: define who tests what — real drivers, real routes, real weather
  - Perform: customer drives in Munich winter, family loads it for a weekend trip, fleet manager checks real fuel savings
- **Enablers** — test driver pool, real-world routes, customer feedback platform
- **Outputs**:
  - Validated SUV — customers confirm it solves their problem
  - Validation report — "real-world EV range: 48km vs 50km target → acceptable"
  - Gaps found → fed back to design team for next generation

## Operation Process

- *The purpose of the Operation Process is to use the system in order to deliver its services.*



## Operation Process: HSUV

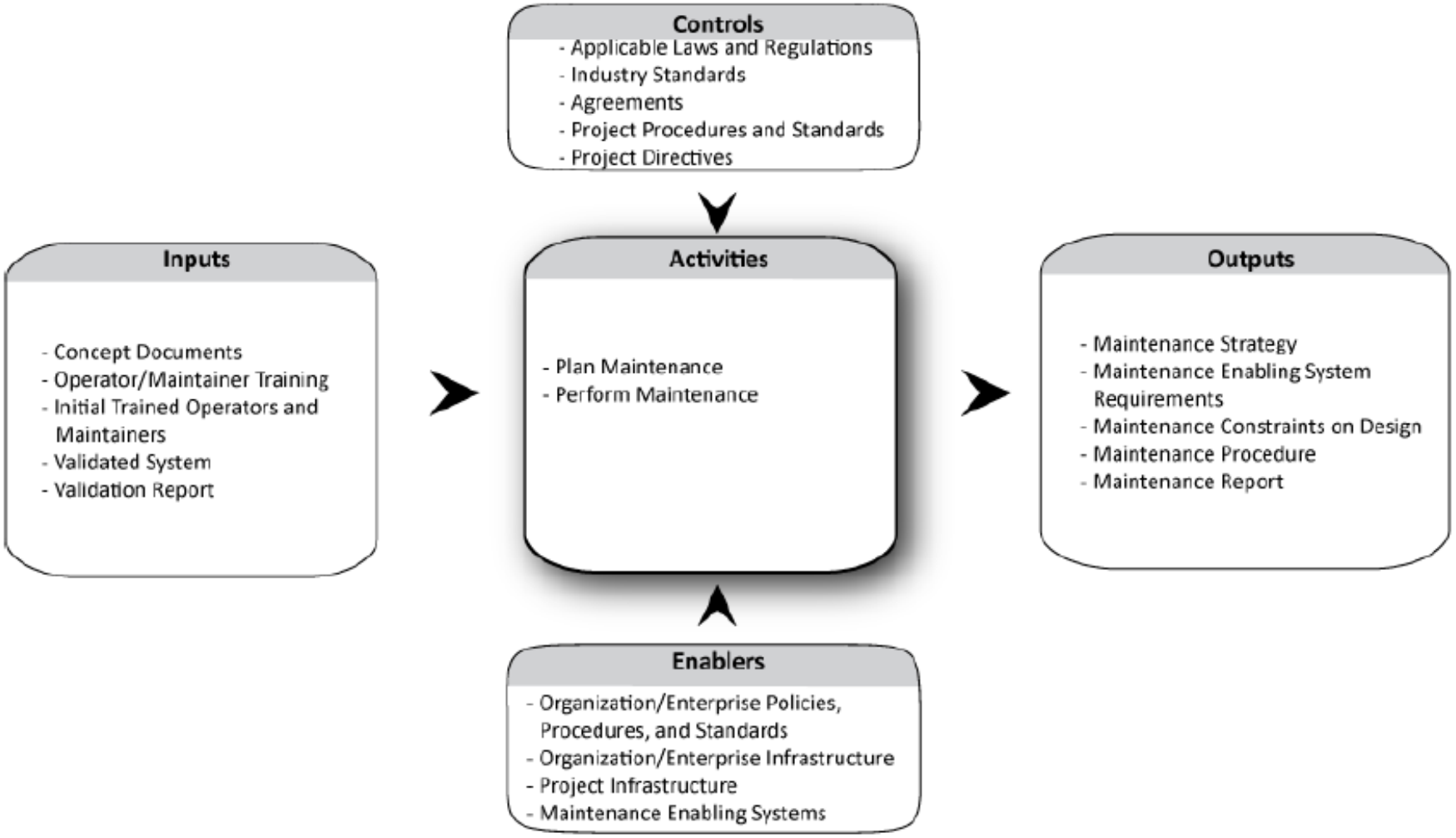
**SUV is with the customer — just needs to do its job every day for 15 years.**

- **Inputs** — validated SUV, validation report, trained technicians
- **Controls** — EU road traffic laws, BMW warranty conditions, emissions regulations
- **Activities**:
  - Prepare: activate car, set up connected services, register owner
  - Operate: customer drives daily — EV mode in city, hybrid on motorway
  - Problem resolution: warning light → BMW remote diagnostics checks battery
  - Support: helpline, app support, OTA software update pushed overnight
- **Enablers** — BMW Connected app, dealer network, remote diagnostics, charging infrastructure
- **Outputs**:
  - SUV delivering its service — moving customer from A to B, saving fuel
  - Operation report — real-world data on battery degradation, range, fault frequency
  - Problems found → flagged back to design for next model

# Technical Process

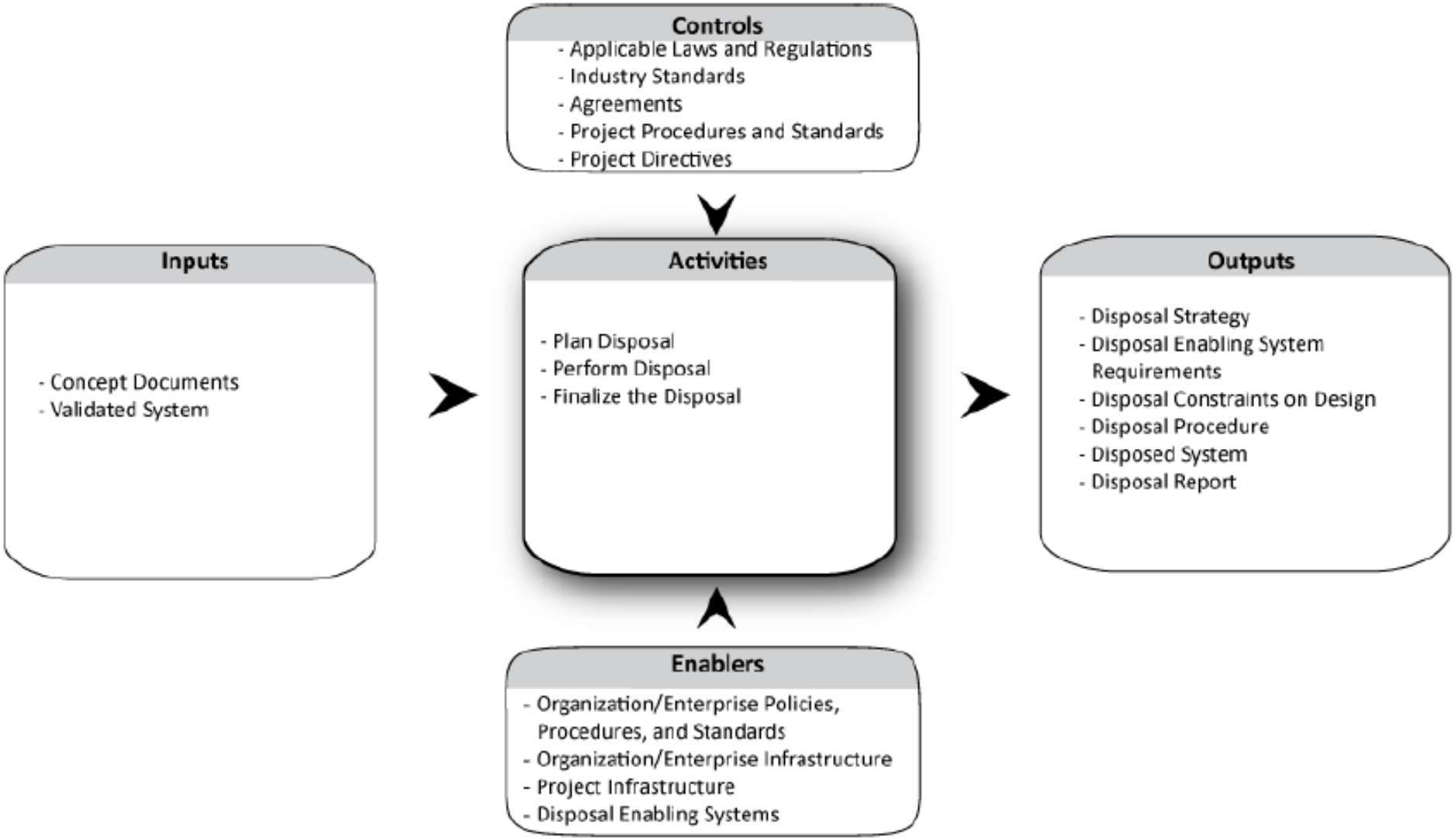
## Maintenance Process

- The SUV is running — now keep it that way. Maintenance ensures the car stays capable of doing its job throughout its entire life.



## Disposal Process

- *The purpose of the Disposal Process is to end the existence of a system entity.*



# Technical Process

## Disposal Process: HSUV

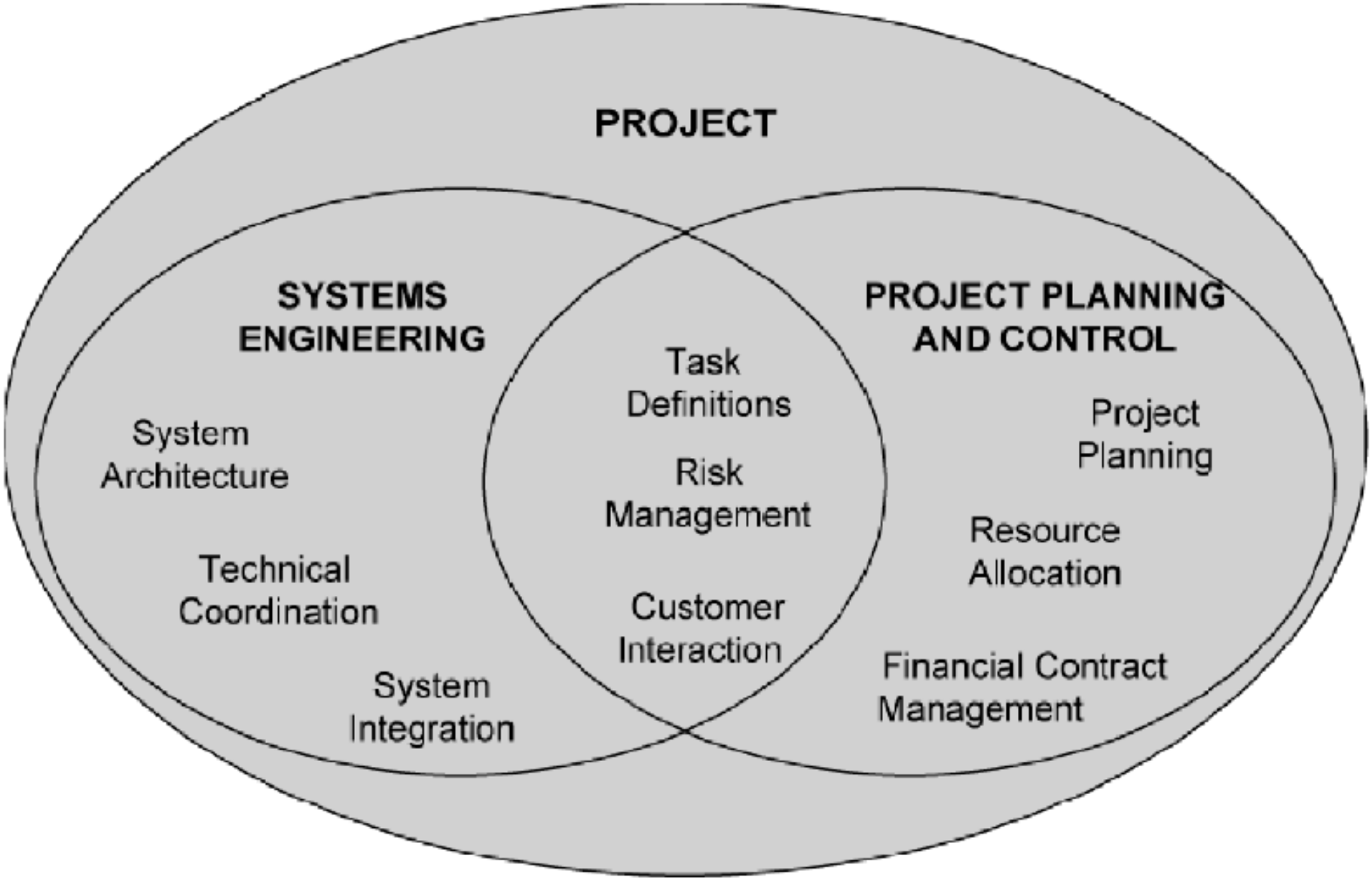
- *The SUV has reached end of life — now retire it responsibly, recover what can be reused, and close everything down cleanly.*
- **Inputs** — maintenance history, material composition documents, decommission plan
- **Controls** — EU End-of-Life Vehicle Directive, battery recycling regulations
- **Activities**:
- Remove and test battery — ≥70% capacity → second life as home energy storage
- Recycle battery → recover lithium, cobalt, nickel

**Enablers** — certified recycling partners, BMW battery second-life programme

- **Outputs**:
- Recovered materials back in supply chain
- Disposal report — proof of compliant end-of-life handling
- Lessons learned → fed to next generation design team

# Project Plans

## Project planning and control / overlap with SE



# II.5. Project Process

- The Project Processes are used to **establish** and **evolve project plans**, to **execute** the **project plans**, to **assess actual achievement** and progress against the plans and to **control execution** of the project through to fulfillment. Individual Project Processes may be invoked at any time in the life cycle and at any level in a hierarchy of projects, as required by project plans or unforeseen events. The Project Processes are applied with a level of rigor and formality that depends on the risk and complexity of the project

- [Agreement] processes define the **activities necessary** to **establish an agreement between two organizations**. If the Acquisition Process is invoked, it provides the means for **conducting business with a supplier**: of products that are supplied for use as an operational system, of services in support of operational activities, or of elements of a system being developed by a project. If the Supply Process is invoked, it provides the means for conducting a project in which the result is a product or service that is delivered to the acquirer.

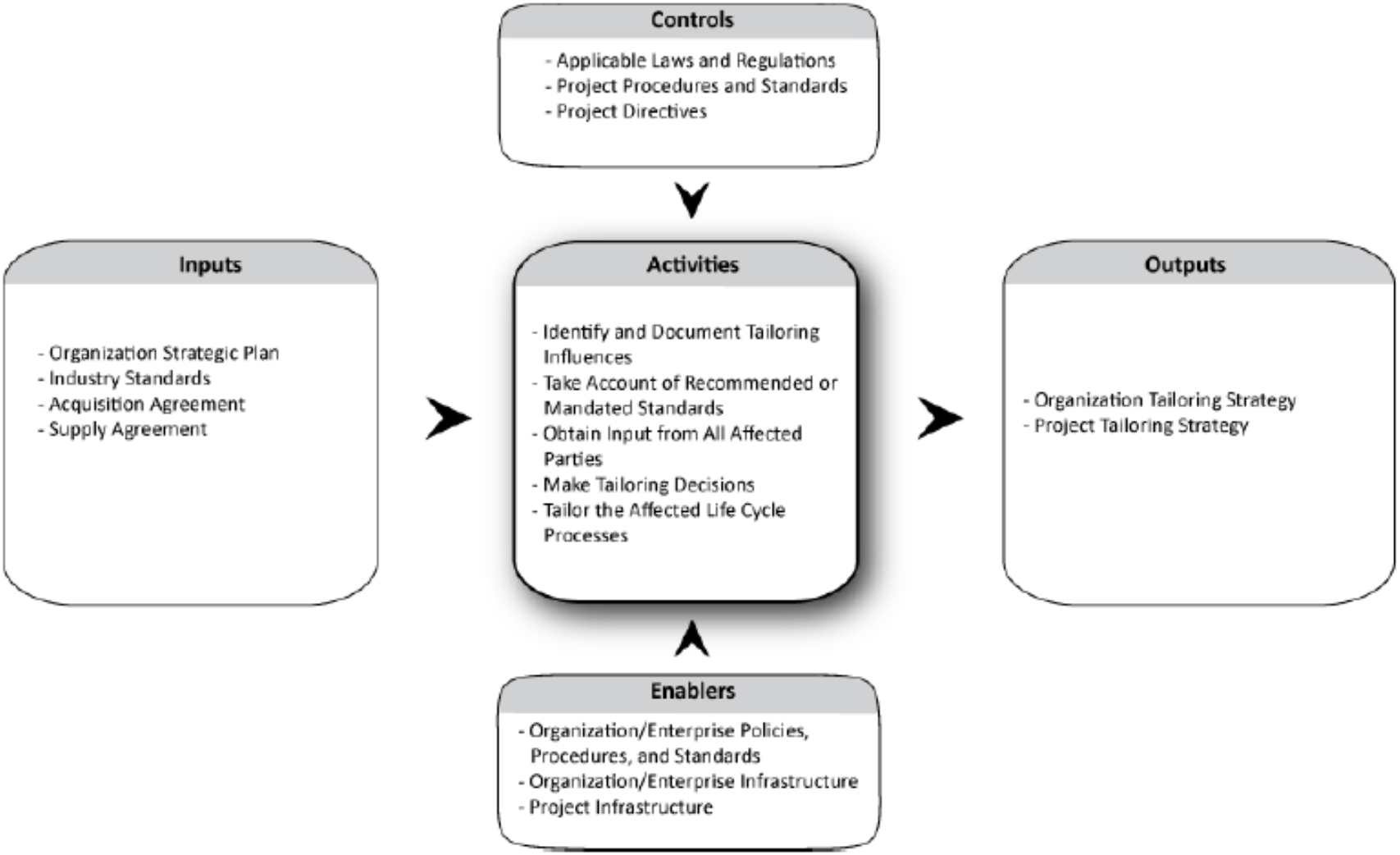
# II.7 Organizational Project Enabling Process

- The Organizational Project-Enabling Processes **ensure** the **organization's capability to acquire and supply products or services** through the initiation, support and control of projects. They **provide resources** and **infrastructure** necessary to support projects and ensure the satisfaction of organizational objectives and established agreements. They are not intended to be a comprehensive set of business processes that enable strategic management of the organization's business.
- Five relevant processes
  - Life Cycle Model Management, Infrastructure Management, Product Portfolio Management, Human Resource Management, and Quality Management.

# II.8 Tailoring Process

- Standards and handbooks are written to address **generic practices** that may or **may not apply to a given organization** or system-of-interest. Most are accompanied by a recommendation to **adapt the processes and activities to the situation at hand**. This adaptation is **called tailoring**

# 51 Tailoring Process



# Tailoring Process: HSUV

ISO 15288 is written for every project on earth — tailoring decides which parts actually apply to *this* SUV.

- **Inputs** — BMW strategic plan, ISO/IEC 15288, supplier agreements
- **Controls** — EU regulations, BMW project directives, mandatory ISO requirements
- **Activities**:
  - Identify what is special about this project
  - Check which standards are mandatory vs optional
  - Consult all teams — engineering, legal, procurement
  - Decide — "full ISO 26262 for battery software, simplified for cup holder"

**Enablers** — BMW process library, project management office, legal team

- **Outputs**:
  - Organisation Tailoring Strategy — BMW's rules for all hybrid projects
  - Project Tailoring Strategy — specific decisions for this SUV only

We discussed systems engineering based on the INCOSE SE  Handbook. Main Processes and activities are addressed. These enables engineering complex systems. For advanced mechanical systems, as we consider in the lecture, not only processes and activities are required in a integrated manner on systems level. It is also required to integrate all processes of all disciplines inclusive the different models.

# Bibliography

[INCOSE2014] INCOSE Systems Engineering Handbook v. 4, October 2014, INCOSE-TP-2003-002-03.2.2